

EUROSTARS

Eurostars (Eureka) SME Innovation Project

AI-Driven Predictive Maintenance for Distributed Wind Turbine Fleets WINDSAGE

Field	Value
Call Identifier	Eurostars Call 2026
Type of Action	Eurostars
Duration	36 months
TRL Range	TRL 3 to TRL 7
Total Budget	EUR 2,400,000
EU Contribution	EUR 1,300,000

Abstract

WINDSAGE develops and validates an edge-plus-cloud predictive maintenance platform for wind turbine fleets that fuses vibration, SCADA, and acoustic-emission sensor data with physics-informed machine learning to predict drivetrain and blade failures four to eight weeks in advance - a lead time substantially exceeding the two-to-four-week horizon of current commercial condition-monitoring systems. The platform's core innovation is the integration of validated reduced-order aeroelastic simulation models as physics priors within a transfer-learning architecture, enabling longer prognostic lead times, lower false-positive rates, and cross-OEM fleet transferability that purely data-driven approaches cannot achieve. A ruggedised, low-power (5 W) edge inference device processes sensor data continuously on-turbine; a scalable multi-tenant cloud SaaS platform aggregates fleet health intelligence and delivers actionable maintenance recommendations through a web dashboard and REST API. The consortium of three complementary SMEs - Nordic Turbine Analytics AS (Norway; project coordinator, ML and cloud software), Ventus Sensors GmbH (Germany; edge hardware), and Delft Dynamics BV (Netherlands; aeroelastic modelling) - contributes precisely the skills required and holds direct access to validation sites through NTA's existing operator relationships. The 36-month project advances the platform from TRL 3 to TRL 7, validated on a fleet of at least 12 turbines across onshore and offshore sites covering three OEM types. Commercial launch is targeted within 12 months of project end. The EUR 2.4M project addresses a EUR 17 billion global wind O&M market growing at 8–10% CAGR, targeting 3,000 paying turbines and EUR 3.6M annual recurring revenue by year three post-launch. WINDSAGE directly supports EU Green Deal objectives by reducing wind energy O&M costs and improving fleet availability for the clean energy transition.

Keywords: predictive maintenance, wind turbine, condition monitoring, physics-informed machine learning, edge computing, aeroelastic modelling, Industry 4.0, SaaS, renewable energy, digital twin

SECTION 1 - PROJECT AND TECHNOLOGY

1.1 Objectives and Ambition

WINDSAGE aims to develop, integrate, and validate a commercially deployable edge-plus-cloud predictive maintenance platform for onshore and offshore wind turbine fleets. The platform will fuse vibration, SCADA, and acoustic-emission sensor data with physics-informed machine-learning models to predict drivetrain and blade failures four to eight weeks in advance - a prognostic lead time that substantially exceeds the two-to-four-week horizon achievable by current commercial condition-monitoring systems. The project pursues three primary technical objectives: (1) design and validate a hybrid physics-informed ML architecture that couples reduced-order aeroelastic simulation outputs with real-time field sensor data, achieving a prognostic lead time of at least four weeks with a false-positive rate below 10% on held-out validation data; (2) develop and qualify a ruggedised, low-power edge inference device capable of continuous on-turbine data pre-processing and anomaly scoring within a 5 W power envelope and to IP67 ingress protection; and (3) integrate edge and cloud components into a multi-tenant SaaS platform supporting fleets of up to 500 turbines per operator account, validated on live turbine data from consortium partners' existing customer sites. These objectives are specific, measurable, achievable within the 36-month timeframe, and directly relevant to EU Green Deal priorities for reducing the levelised cost of wind energy.

The ambition of WINDSAGE decisively surpasses the current state of practice. Existing commercial condition-monitoring solutions rely primarily on vibration analysis using spectral or shallow-ML statistical methods without physics priors, delivering only short-horizon warnings with elevated false-positive rates. WINDSAGE embeds aeroelastic domain knowledge directly into the model architecture, enabling long-horizon prognosis under variable wind and load regimes that purely data-driven approaches cannot reliably extrapolate - particularly when labelled failure data are scarce. This constitutes a qualitative architectural advance, not merely an incremental improvement to existing techniques, and positions WINDSAGE to meet the growing demand for advanced O&M intelligence as European wind fleets age and expand offshore.

1.2 Relation to the Work Programme

The state of the art in wind turbine condition monitoring is characterised by three technology families: (i) vibration-based CMS employing ISO 13373 spectral analysis, deployed commercially by vendors such as SKF, Schaeffler, and Brüel & Kjær; (ii) SCADA-analytics platforms using statistical process control and shallow anomaly detection, offered by incumbents including DNV Windmaster, Romax Spectrum, and Greenbyte; and (iii) emerging data-driven ML platforms (e.g., Uptake, SparkCognition) that apply deep learning to SCADA streams but lack physics grounding and require large, turbine-type-specific labelled failure datasets. None of these approaches integrates aeroelastic simulation as a generative prior for failure prognosis. This is the critical gap that WINDSAGE addresses.

The technical challenges involved in closing this gap are threefold. First, aeroelastic models are computationally expensive and have traditionally been used offline for design-phase analysis rather than in real-time inference loops; making them tractable for continuous operational prognosis requires the development of validated reduced-order models (ROMs). Second, fusing heterogeneous sensor modalities - high-frequency vibration, acoustic emission, and low-frequency SCADA - demands robust multi-rate data alignment and feature extraction pipelines. Third, deploying inference at the turbine edge under strict power and connectivity constraints requires novel model compression and hardware co-design techniques not yet available as off-the-shelf solutions. WINDSAGE addresses all three challenges in an integrated programme, differentiating from all identified competitors through longer prognostic lead times, lower false-positive rates, and transferability across turbine makes and models - a decisive commercial advantage for independent

service providers managing heterogeneous fleets.

1.3 Concept and Methodology

The WINDSAGE technical approach is structured around three tightly integrated layers, developed in parallel across WP2–WP4 and converging in the WP5 field validation.

The first layer is the physics-informed ML core. Delft Dynamics BV will develop validated reduced-order aeroelastic models (ROMs) of representative drivetrain and blade substructures using multi-body and finite-element methods. These ROMs generate physics-consistent synthetic failure trajectories that are used to pre-train the prognostic neural network, directly overcoming the scarcity of real failure labels that limits all purely data-driven approaches. Nordic Turbine Analytics AS will then fine-tune the pre-trained model on field data using a transfer-learning protocol, adapting to site- and turbine-specific operating conditions without requiring retraining from scratch for each new asset.

The second layer is the edge inference device. Ventus Sensors GmbH will design a ruggedised edge module integrating a multi-channel vibration and acoustic-emission acquisition front-end with a low-power AI inference chip. Model compression techniques - specifically structured pruning and 8-bit quantisation - will be applied to deploy the WINDSAGE prognostic model within the 5 W power envelope, enabling continuous on-turbine inference with minimal connectivity dependence.

The third layer is the cloud SaaS platform. Nordic Turbine Analytics AS will extend its existing analytics platform to ingest edge-processed feature vectors, aggregate fleet-level health indices, and present actionable maintenance recommendations through web and API interfaces. The platform will support multi-tenant operation for fleets of up to 500 turbines per operator account and will integrate with common CMMS systems via a REST API.

The three layers are developed through an iterative hardware-software co-development cycle, with integration checkpoints at MS3 (ML model ready) and MS4 (edge device qualified) before the WP5 field validation commences. The 12-turbine field validation covers both onshore and offshore assets across at least three turbine OEM types, providing a rigorous and representative test of system performance.

Key technical risks and their mitigations are as follows: model transferability across turbine OEM types is mitigated by the ROM-based pre-training architecture and by explicit turbine-type embeddings in the transfer-learning protocol; edge hardware certification timelines are mitigated by maintaining a parallel software-only (cloud-inference) deployment path throughout WP3; and data access from operator sites is mitigated by existing data-sharing agreements within NTA's customer base, supported by letters of intent secured before project start.

1.4 Interdisciplinary Considerations

WINDSAGE is inherently interdisciplinary, integrating structural dynamics and aeroelastic engineering (Delft Dynamics BV), embedded systems and industrial sensing hardware (Ventus Sensors GmbH), and cloud software engineering with applied machine learning (Nordic Turbine Analytics AS). The project synthesises knowledge from mechanical engineering, signal processing, AI/ML, and industrial IoT in a configuration that no single partner could achieve independently, and that has no direct precedent in the published literature on wind turbine condition monitoring.

Open Science principles are observed throughout. All partners commit to publishing project methodology in open-access peer-reviewed journals following IP protection (patent filing by month 24). Benchmark datasets generated during pre-training and validation will be released to the wind energy research community, subject to operator consent and GDPR-compliant anonymisation. A Data

Management Plan (D1.1) will be submitted by month 3, in line with Eurostars and national funder requirements.

Gender balance is actively promoted: all consortium partners commit to applying gender-inclusive recruitment practices for all project roles and to monitoring gender representation in the project team at each annual review. The project's subject matter - sensor data analytics for industrial equipment - raises no significant ethical concerns beyond standard data privacy considerations for operational sensor data. These are addressed through GDPR-compliant data processing agreements with all operator validation sites, documented in the Data Management Plan. The project has an exclusively civilian purpose.

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SECTION 2 - MARKET AND IMPACT

2.1 Pathways Towards Impact

The global wind operations and maintenance (O&M) market is valued at approximately EUR 17 billion annually and is forecast to grow at 8–10% CAGR through 2030, driven by fleet ageing and the rapid expansion of offshore wind capacity. Unplanned downtime accounts for an estimated 30–40% of total O&M cost for wind operators, with drivetrain and blade failures being the dominant contributors. WINDSAGE targets two primary customer segments: wind farm operators (utilities and independent power producers) seeking to reduce unplanned downtime and extend asset life, and independent service providers (ISPs) managing third-party heterogeneous fleets who require a turbine-agnostic platform. Together, these segments encompass over 400,000 utility-scale turbines in Europe alone.

The WINDSAGE platform will be commercialised by Nordic Turbine Analytics AS under a SaaS subscription model priced at EUR 800–1,200 per turbine per year, complemented by a one-time hardware fee for the Ventus edge module (target price EUR 2,500–3,500 per turbine). This pricing positions WINDSAGE well within the payback thresholds of target customers, given that a single avoided unplanned gearbox failure typically saves EUR 150,000–300,000 in parts, crane mobilisation, and lost generation.

Go-to-market strategy leverages NTA's existing relationships with five European wind farm operators covering approximately 1,200 turbines, providing a direct route to first commercial contracts within 12 months of project end. Initial market entry will focus on Germany and the Netherlands, supported by Ventus Sensors GmbH and Delft Dynamics BV through their respective industry networks, before expanding to the broader European and Nordic markets. The project targets commercial launch by month 38 (two months after project close), with 500 paying turbines under contract in year one post-launch and 3,000 turbines by year three - generating approximately EUR 3.6M annual recurring revenue at mid-range subscription pricing.

At the systemic level, widespread adoption of WINDSAGE-class prognostic maintenance will contribute to EU energy policy objectives by reducing wind energy O&M costs, improving fleet availability, and supporting the European Green Deal target of 510 GW of installed wind capacity by 2030. The field validation results (targeted: 40% reduction in unplanned downtime versus pre-deployment baseline) will be published in a public report to maximise knowledge transfer to the broader wind industry.

2.2 Dissemination, Exploitation and Communication

Exploitation rights follow task ownership, as will be formalised in the Consortium Agreement signed before project start. The hybrid physics-informed ML algorithms, pre-trained model weights, and cloud SaaS platform are owned by Nordic Turbine Analytics AS. The edge acquisition and inference hardware design, firmware, and associated manufacturing know-how are owned by Ventus Sensors GmbH. The reduced-order aeroelastic model library is owned by Delft Dynamics BV, which grants NTA a royalty-free, perpetual licence for integration within the WINDSAGE platform; Delft Dynamics retains the right to license the ROM library independently to other wind energy clients as a consulting asset.

To protect the consortium's competitive position, NTA will file at least one patent application covering the physics-informed transfer-learning architecture no later than month 24. Ventus Sensors GmbH will initiate the CE marking process for the edge device under the applicable Machinery Directive and Radio Equipment Directive, with the objective of achieving market-ready certification in parallel with the WP5 validation campaign. Trade secret protection will be applied to training

datasets and proprietary model configurations not covered by the patent.

All partners will publish project methodology papers in open-access venues (targeting journals such as Renewable Energy, Mechanical Systems and Signal Processing, and IEEE Transactions on Industrial Electronics) after IP protection is secured, in compliance with Eurostars open-access requirements. Benchmark datasets will be released under a Creative Commons licence where operator consent is obtained.

Communication and dissemination activities, coordinated under WP1 (T1.2), will include: presentations at WindEurope Annual Event and WindEnergy Hamburg; articles in trade publications such as Windpower Monthly and EWEA Offshore; a project website updated at key milestones; and joint press releases at project kick-off, MS4 (edge device qualification), and MS6 (TRL 7 validation confirmed). These activities are designed to build market awareness ahead of the commercial launch and to attract additional early-adopter operator relationships.

2.3 Summary of Impact Indicators

Indicator	Target	Measurement
Cumulative paying turbines under WINDSAGE SaaS contract	3,000 turbines by end of year 3 post-launch	CRM subscription records, annual report
Revenue generated by NTA from WINDSAGE platform	EUR 3.6M ARR by year 3 post-launch	NTA audited financial statements
New full-time jobs created across consortium	12 FTE by year 3 post-launch (NTA: 7, Ventus: 3, Delft Dynamics: 2)	HR records reported at annual review
Reduction in unplanned downtime for validated operator sites	40% reduction versus pre-deployment baseline	Operator SCADA availability logs, field validation report (WP5)
Patent applications filed	At least 1 by month 24	Patent office filing receipts
Time to market from project end	Commercial launch within 12 months of project end (month 48 from project start)	First commercial contract signed

SECTION 3 - CONSORTIUM AND IMPLEMENTATION

3.1 Work Plan and Work Packages

WP	Title	Lead	PM	Duration
WP1	Project Management and Dissemination	Nordic Turbine Analytics AS	9	M1-M36
WP2	Physics-Informed ML Architecture and Reduced-Order Models	Delft Dynamics BV	38	M1-M20
WP3	Edge Device Development and Integration	Ventus Sensors GmbH	42	M4-M28
WP4	Cloud SaaS Platform Development	Nordic Turbine Analytics AS	44	M6-M30
WP5	Field Validation and Demonstration	Nordic Turbine Analytics AS	25	M24-M36

WP1: Project Management and Dissemination

Lead: Nordic Turbine Analytics AS | Person-months: 9 | Duration: M1 - M36

WP1 provides the governance, quality assurance, financial management, and communication infrastructure required for the effective execution of the WINDSAGE project. Nordic Turbine Analytics AS, as project coordinator, will operate a lean project management office responsible for maintaining the project schedule, risk register, and quality plan; convening quarterly consortium meetings and annual review meetings; and preparing interim and final reports for submission to national programme coordinators (NPCs). A Consortium Agreement will be finalised and signed before project start, establishing IP ownership, liability provisions, and decision-making procedures. A dissemination and communication plan will be delivered by month 2 and executed continuously, ensuring that scientific, industrial, and policy audiences are reached at appropriate project stages. WP1 runs for the full 36-month project duration and interfaces with all other WPs through formal review gates aligned with the milestone schedule.

T1.1: Project coordination and governance - Establish and maintain consortium governance structures, including the Steering Committee and Technical Advisory Board. Maintain and update the project risk register, quality plan, and financial monitoring reports on a quarterly basis. Prepare and submit interim progress reports at month 18 and a final report at month 36, in the format required by Eurostars NPCs. Serve as the single point of contact between the consortium and national funding bodies.

T1.2: Dissemination and communication - Develop and execute the dissemination and communication plan (month 2). Activities include: maintaining a public project website; preparing and submitting conference abstracts and papers for WindEurope, WindEnergy Hamburg, and peer-reviewed open-access journals; issuing joint press releases at key milestones; and publishing trade press articles in Windpower Monthly and equivalent outlets. Ensure all open-access publication obligations are met in compliance with Eurostars and national funder requirements.

WP2: Physics-Informed ML Architecture and Reduced-Order Models

Lead: Delft Dynamics BV | Person-months: 38 | Duration: M1 - M20

WP2 develops the scientific and technical core of the WINDSAGE prognostic engine: validated reduced-order aeroelastic models (ROMs) for representative drivetrain and blade substructures, and a hybrid physics-informed ML architecture that integrates ROM-generated failure trajectories as physics priors. Delft Dynamics BV leads ROM development, leveraging its established capability in multi-body and finite-element wind turbine modelling. Nordic Turbine Analytics AS leads the ML architecture design and the pre-training and transfer-learning protocol. WP2 runs from month 1 to month 20 and its primary outputs - the validated ROM library (D2.1) and the pre-trained prognostic model specification (D2.2) - feed directly and critically into WP3 (model compression and edge deployment) and WP4 (cloud platform integration). The success criterion for WP2 is a prognostic model demonstrating a lead time of at least four weeks with RMSE below 5% versus full-order model predictions on held-out historical datasets, confirmed at MS3.

T2.1: Reduced-order aeroelastic model development - Develop and validate a ROM library covering the dominant failure modes of the gearbox, main bearing, and blade structure, using a combination of multi-body dynamics and finite-element methods calibrated against publicly available reference datasets and, where possible, Delft Dynamics BV's proprietary simulation archive. Validation criterion: ROM prediction error (RMSE) below 5% relative to full-order model outputs across representative load cases. Deliver ROM library v1.0 as D2.1 at month 12.

T2.2: Physics-informed ML architecture design - Design a hybrid neural network architecture - combining convolutional feature extraction from sensor time series with a physics-consistent decoder conditioned on ROM-generated failure trajectory embeddings - for remaining-useful-life (RUL) prognosis of drivetrain and blade components. Architecture design will be guided by a structured literature review and benchmarking against baseline LSTM and autoencoder approaches on historical field datasets provided by NTA operator partners.

T2.3: Pre-training and synthetic data generation - Use the validated ROM library to generate a large-scale synthetic failure dataset spanning diverse wind conditions, load regimes, and fault severity levels. Pre-train the WP2 ML architecture on the synthetic dataset, then validate the pre-trained model on historical held-out field data to confirm the four-week prognostic lead time and sub-10% false-positive rate targets. Document the pre-training protocol, model hyperparameters, and benchmark results in D2.2.

WP3: Edge Device Development and Integration

Lead: Ventus Sensors GmbH | Person-months: 42 | Duration: M4 - M28

WP3 develops the ruggedised edge acquisition and inference module that is a hardware-differentiating component of the WINDSAGE system. Ventus Sensors GmbH leads all hardware development activities, from architecture specification through prototyping and qualification testing. Nordic Turbine Analytics AS contributes to the edge-cloud interface specification and validates end-to-end data pipeline performance. The edge device must satisfy the following verified performance requirements: 5 W maximum power consumption during continuous inference, IP67 ingress protection, IEC 61400-25 compliant data communication, and support for simultaneous acquisition of at least eight vibration channels and two acoustic-emission channels at up to 51.2 kHz sampling rate. WP3 runs from month 4 to month 28; the qualified Rev B prototype (MS4, month 22) is the prerequisite for edge device installation in the WP5 field validation. A parallel software-only (cloud-inference) deployment path is maintained throughout WP3 as a risk mitigation measure, ensuring that WP5 can proceed even if hardware qualification is delayed.

T3.1: Edge hardware architecture and prototyping - Define the hardware architecture for the edge module, including selection of the multi-channel vibration and acoustic-emission acquisition front-end, AI inference chip (evaluated options include NXP i.MX 8, Hailo-8, and equivalent), power management subsystem, and industrial-grade enclosure. Produce Rev A hardware prototypes and conduct initial functional validation against the power and communication specifications. Incorporate findings into the Rev B design for qualification testing.

T3.2: Model compression and edge firmware - Apply structured pruning and post-training 8-bit quantisation to the WP2 prognostic model to meet the memory and compute constraints of the selected edge inference chip. Target: model inference latency below 100 ms per turbine measurement epoch with less than 2% degradation in prognostic accuracy relative to the full-precision cloud model. Develop edge firmware for sensor acquisition, on-device pre-processing (feature extraction, anomaly scoring), and secure data packaging for transmission to the cloud platform.

T3.3: Edge-cloud integration and communication protocol - Implement a secure MQTT over TLS data pipeline for transmission of compressed feature vectors and anomaly scores from the edge device to the WINDSAGE cloud platform (WP4). Validate end-to-end pipeline latency, data integrity, and reconnection behaviour under simulated low-bandwidth and intermittent-connectivity conditions representative of remote wind farm sites. Document integration test results in D3.2.

WP4: Cloud SaaS Platform Development

Lead: Nordic Turbine Analytics AS | Person-months: 44 | Duration: M6 - M30

WP4 develops the multi-tenant cloud SaaS platform that aggregates edge-processed data and raw SCADA streams, runs fleet-level health intelligence, and presents actionable maintenance recommendations to operators and ISPs. Nordic Turbine Analytics AS leads all software development, building on its existing analytics platform infrastructure to accelerate development and reduce technical risk. Ventus Sensors GmbH provides detailed edge data format specifications to ensure ingestion pipeline compatibility; Delft Dynamics BV provides model update protocols to support ROM-informed recalibration of the prognostic engine as new field data accumulate. WP4 runs from month 6 to month 30, delivering a Beta platform release (D4.1) at month 28, in time for deployment at the WP5 validation sites. Platform scalability is designed from the outset for fleets of up to 500

turbines per operator account, with a cloud architecture capable of supporting at least ten simultaneous operator tenants.

T4.1: Platform architecture and data ingestion pipeline - Design a scalable, cloud-native platform architecture (targeting Kubernetes-based microservices deployment on a major cloud provider) capable of multi-tenant operation. Implement the data ingestion pipeline for edge-processed feature vectors arriving via the MQTT/TLS pipeline (WP3) and for raw SCADA data streams ingested via direct API connections to operator SCADA historians. Implement data quality monitoring and automated gap-filling procedures to handle real-world data availability issues.

T4.2: Fleet health dashboard and alerting engine - Develop a web-based operator dashboard presenting per-turbine and fleet-level health indices, prognostic remaining-useful-life timelines, and root-cause indicators for flagged anomalies. Implement a configurable alerting engine with user-defined threshold settings, alert routing (email, SMS, API webhook), and a feedback mechanism allowing operators to log maintenance actions against alerts - providing a supervised labelling loop to improve model performance post-deployment.

T4.3: API and CMMS integration - Design and implement a documented REST API exposing WINDSAGE health indices, prognostic timelines, and maintenance recommendations to third-party systems. Validate integration with at least one reference CMMS platform (SAP PM or IBM Maximo) in collaboration with a reference customer from NTA's existing operator network. Publish the API specification as D4.2 (public dissemination level) to facilitate adoption by additional CMMS vendors and customers post-project.

WP5: Field Validation and Demonstration

Lead: Nordic Turbine Analytics AS | Person-months: 25 | Duration: M24 - M36

WP5 validates the integrated WINDSAGE system at TRL 7 - demonstration in an operational environment - on a fleet of at least 12 operational turbines across a minimum of two sites, covering both onshore and offshore assets and at least three turbine OEM types. The validation protocol is defined in D5.1 (month 25) and covers three primary KPIs: (i) prognostic lead time for identified fault events (target: ≥ 4 weeks mean lead time); (ii) false-positive rate for maintenance alerts (target: $< 10\%$ of total alerts); and (iii) reduction in unplanned downtime relative to the 12-month pre-deployment baseline (target: $\geq 40\%$). The 10-month continuous monitoring campaign (T5.2) produces the ground-truth dataset against which all KPIs are assessed. Results are analysed in T5.3 and published in a public validation report (D5.2), providing independent evidence of system efficacy for prospective customers and contributing to the broader wind energy research community.

T5.1: Field site preparation and system deployment - Coordinate with validation site operators to finalise data-sharing agreements, install edge devices on selected turbines, configure cloud platform tenant accounts, and establish pre-deployment baseline SCADA performance metrics for each turbine. Complete all health and safety, access, and cybersecurity requirements for each site. Deliver validated deployment checklist and confirm seven-day continuous system operation as verification of M55.

T5.2: Validation campaign and data collection - Execute a 10-month continuous monitoring campaign across all validation turbines, collecting prognostic performance data (health indices, alert events, prognostic lead times) alongside ground-truth maintenance records from operator CMMS systems. Monitor data quality and system uptime weekly; perform model recalibration updates as needed using the online transfer-learning protocol developed in WP2. Cover at least three turbine OEM types to demonstrate cross-fleet transferability.

T5.3: Validation analysis and final reporting - Conduct a comprehensive statistical analysis of validation campaign data to quantify all three primary KPIs against pre-defined targets. Assess prognostic performance by OEM type, site type (onshore/offshore), and fault mode category. Produce the public field validation report (D5.2) documenting methodology, results, and limitations. Compile and submit the final project report for NPC review, including evidence of TRL 7 achievement and a commercialisation readiness assessment.

3.2 Consortium Composition

Organisation	Country	Type	Role	Description
Nordic Turbine Analytics AS	Norway	SME	Coordinator	Project coordinator; lead developer of physics-informed ML algorithms, cloud SaaS platform, and field validation; primary route to market via existing operator

				relationships.
Ventus Sensors GmbH	Germany	SME	Partner	Lead developer of ruggedised edge acquisition and inference hardware; responsible for model compression, firmware, and CE marking pathway.
Delft Dynamics BV	Netherlands	SME	Partner	Lead developer of reduced-order aeroelastic models and physics-informed ML architecture; provides domain expertise in structural dynamics and wind turbine loads.

The WINDSAGE consortium brings together three highly complementary SMEs from three Eurostars-eligible countries - Norway (Nordic Turbine Analytics AS), Germany (Ventus Sensors GmbH), and the Netherlands (Delft Dynamics BV) - each contributing a distinct, non-overlapping technical capability that is individually necessary but collectively sufficient for the project's success. No single partner, nor any credible combination of two, could deliver the integrated system that is WINDSAGE's core innovation and commercial proposition.

Nordic Turbine Analytics AS contributes cloud software platform expertise, applied machine learning capability, and direct commercial access to European wind farm operators through an existing customer base of approximately 1,200 turbines - the critical route to market and to the field validation data that the project depends upon. Ventus Sensors GmbH contributes embedded hardware engineering, industrial sensing design, and manufacturing pathway expertise for the ruggedised edge inference module; these capabilities are entirely absent from the other two partners. Delft Dynamics BV contributes deep aeroelastic modelling expertise and a validated multi-body/FE simulation toolchain; this physics domain knowledge is the generative foundation of the WINDSAGE approach and cannot be replicated by a software or hardware company within the project timeframe.

The transnational collaboration is not merely additive: the system's competitive differentiation - longer prognostic lead times and cross-OEM transferability - is inseparable from the integration of Delft Dynamics BV's physics priors into NTA's ML architecture, deployed on Ventus Sensors GmbH's edge hardware. Each partner also realises clear individual commercial benefit: NTA gains a validated, differentiated product for international scale-up; Ventus Sensors GmbH gains a new wind energy hardware product line with a ready route to market through NTA; and Delft Dynamics BV gains a licensed IP asset and expanded credentials in data-driven wind energy services.

The budget distribution is consistent with Eurostars eligibility rules: no single country exceeds 70% of the total project budget, and the SME share of the budget (excluding subcontracting, which is zero) exceeds 50% of the total project cost. All three partners are independent SMEs meeting the Eurostars SME definition.

3.3 Milestones

ID	Title	WP	Due	Verification
MS1	Consortium Agreement signed and project kick-off	WP1	M2	Signed Consortium Agreement submitted to NPC;

	completed			kick-off meeting minutes circulated
MS2	Reduced-order aeroelastic model library validated against reference datasets	WP2	M12	Validation report with ROM accuracy metrics (RMSE < 5% vs full-order model) approved by consortium
MS3	Physics-informed ML model pre-trained and benchmarked on historical field data	WP2	M20	Benchmark report demonstrating prognostic lead time ≥ 4 weeks on held-out dataset
MS4	Edge device prototype (Rev B) meeting power and communication specifications	WP3	M22	Hardware qualification test report signed off by Ventus engineering team
MS5	Integrated edge-cloud system deployed at first validation site	WP5	M26	Deployment checklist completed; system operational log showing 7-day continuous operation
MS6	Field validation completed; TRL 7 confirmed	WP5	M35	Validation report with KPI results; independent review by NPC if required

3.4 Deliverables

ID	Title	WP	Due	Type	Dissemination
D1.1	Data Management Plan	WP1	M3	Report	Public
D1.2	Final Project Report and Dissemination Summary	WP1	M36	Report	Public
D2.1	Reduced-Order Aeroelastic Model Library (v1.0)	WP2	M12	Software/Dataset	Confidential
D2.2	Physics-Informed ML Architecture Specification and Pre-trained Model	WP2	M20	Software	Confidential
D3.1	Edge Device Hardware Design Package (schematics, BOM, firmware)	WP3	M22	Prototype/Technical Document	Confidential
D3.2	Edge-Cloud Integration Test Report	WP3	M28	Report	Confidential
D4.1	WINDSAGE Cloud SaaS Platform (Beta Release)	WP4	M28	Software	Confidential

D4.2	API and CMMS Integration Specification	WP4	M30	Technical Document	Public
D5.1	Field Validation Protocol	WP5	M25	Report	Confidential
D5.2	Field Validation Report (public summary)	WP5	M36	Report	Public

3.5 Resources and Budget Summary

Cost Category	Amount (EUR)
Personnel Costs	1680000
Subcontracting	0
Travel & Subsistence	72000
Equipment	180000
Other Direct Costs	108000
Indirect Costs (25%)	360000
TOTAL	2400000

3.6 Risk Management

ID	Risk	Likelihood	Impact	Mitigation
R1	Insufficient labelled failure data from operator sites for model fine-tuning	Medium	High	ROM-based synthetic pre-training reduces dependency on real failure labels; existing NTA operator agreements guarantee access to historical SCADA datasets for at least four turbine types.
R2	Edge device fails to meet 5 W power budget with full prognostic model deployed	Medium	Medium	Parallel software-only deployment path (cloud-only inference) maintained throughout WP3; aggressive model compression schedule with go/no-go gate at MS4.
R3	Model transferability degraded across different turbine OEM types	Medium	High	Transfer-learning architecture with turbine-type embeddings designed explicitly for cross-OEM generalisation; validation dataset covers at least three OEM types in WP5.
R4	Operator site access delayed or withdrawn for field validation	Low	High	Letters of intent from two operator sites secured before project start; alternative backup site identified through Ventus Sensors GmbH's

				German customer base.
R5	Key personnel departure from a partner SME	Low	Medium	Critical technical knowledge documented in deliverables at each WP gate; consortium agreement includes personnel continuity provisions and cross-training obligations.
R6	Competitive products reach market with comparable lead times during project execution	Low	Medium	WINDSAGE's physics-informed architecture provides a defensible technical moat; patent filing by month 24 protects core IP; speed-to-market plan targets commercial launch within 12 months of project end.